

CHECKLIST: JAVA GUI ORDERING SYSTEM

PROGRAMMER: [DAN VANICE D. SAMSON](#)

NOTE: Once specific task is completed, record the date on the right side and highlight it in blue. Use Rubric number 2.

1. LOGIN MODULE

USER INTERFACE

- ☒ Create login window (JFrame/JavaFX Stage) - [April 07, 2026](#)
- ☒ Add:
 - ☒ Username field – [April 06, 2026](#)
 - ☒ Password field – [April 06, 2026](#)
 - ☒ Correct Labelling/Labels – [April 06, 2026](#)
 - ☒ Login button – [April 06, 2026](#)
 - ☒ Clear button – [April 06, 2026](#)
 - ☒ Show/Hide password toggle – [April 13, 2026](#)
 - ☒ Register New User / Create an Account – [April 13, 2026](#)
 - ☒ Forgot Password -[April 13, 2026](#)
 - ☒ Three Security Question in Password Recovery - [April 13, 2026](#)
- ☒ Center layout and apply clean design [April 13, 2026](#)
- ☒ Add system title/logo - [April 13, 2026](#)

FUNCTIONALITY

- ☒ Capture input values [April 20, 2026](#)
- ☒ Apply input validation in all Text Fileds [April 20, 2026](#)
- ☒ Validate:
 - ☒ Empty fields (provide error message) [April 20, 2026](#)
 - ☒ Input format (provide error message if incorrect data format) [April 20, 2026](#)
- ☒ Display inline error messages [April 20, 2026](#)
- ☒ Clear all fields once Done [April 20, 2026](#)
- ☒ Focus cursor in textfield1 always [April 20, 2026](#)

DATABASE

- ☒ Connect via JDBC [April 20, 2026](#)
- ☒ Use PreparedStatement [April 20, 2026](#)
- ☒ Query user credentials [April 20, 2026](#)

SECURITY

- ☒ Hash passwords (SHA-256/BCrypt) [April 20, 2026](#)
- ☒ Avoid plain text comparison [April 20, 2026](#)

NAVIGATION

- ☒ Redirect to dashboard on success [April 20, 2026](#)
- ☒ Store session (username/role) [April 20, 2026](#)
- ☒ Dispose all not needed Forms [April 20, 2026](#)

2. DASHBOARD MODULE

USER INTERFACE

- ☒ Create main dashboard window [April 20, 2026](#)
- ☒ Sidebar navigation:
 - ☒ Orders - [April 25, 2026](#)
 - ☒ Products - [April 25, 2026](#)
 - ☒ Customers - [April 25, 2026](#)
 - ☒ Reports - [April 25, 2026](#)
 - ☒ Logout - [April 25, 2026](#)
- ☒ Main content panel (CardLayout) – (I used JTabbedPane for less memory consumption) [April 25, 2026](#)
- ☒ Welcome message - [April 25, 2026](#)

FUNCTIONALITY

- ☒ Add button event handlers - [April 25, 2026](#)
- ☒ Load modules dynamically - [April 25, 2026](#)
- ☒ Display date/time - [April 25, 2026](#)
- ☒ Should have a Calendar upon Click - [April 25, 2026](#)

LOGOUT

- ☒ Confirm logout (provide information/confirmation message) - [April 25, 2026](#)
- ☒ Clear session - [April 25, 2026](#)
- ☒ Return to login - [April 25, 2026](#)

3. ORDER MANAGEMENT MODULE

USER INTERFACE

- ☒ Product list (JTable/ComboBox) - [April 25, 2026](#)
- ☒ Quantity field - [April 25, 2026](#)
- ☒ Add to Cart button - [April 25, 2026](#)
- ☒ Cart table - [April 25, 2026](#)
- ☒ Total price label - [April 25, 2026](#)
- ☒ Place Order button - [April 25, 2026](#)

FUNCTIONALITY

- ☒ Load products from Table or in DB - [April 25, 2026](#)
- ☒ Add item to cart - [April 25, 2026](#)
- ☒ Update quantity if duplicate - [April 25, 2026](#)
- ☒ Remove/edit cart items - [April 25, 2026](#)
- ☒ Auto-compute total - [April 25, 2026](#)

DATABASE

- ☒ Insert into orders table - [April 25, 2026](#)
- ☒ Insert into order details table - [April 25, 2026](#)

EXTRA

- ☒ Generate order ID - [April 25, 2026](#)
- ☒ Reduce stock - [April 25, 2026](#)
- ☒ Show confirmation/receipt - [April 25, 2026](#)
- ☒ Printed receipt - [April 25, 2026](#)

4. PRODUCT MANAGEMENT MODULE

USER INTERFACE

- ☒ Product form:
 - ☒ Name [April 27, 2026](#)
 - ☒ Price [April 27, 2026](#)
 - ☒ Stock [April 27, 2026](#)
- ☒ Product table [April 27, 2026](#)
- ☒ Buttons (*password protected*):
 - ☒ Add [April 27, 2026](#)
 - ☒ Update [April 27, 2026](#)
 - ☒ Delete [April 27, 2026](#)
 - ☒ Clear [April 27, 2026](#)

FUNCTIONALITY

- ☒ Validate inputs [April 27, 2026](#)
- ☒ Populate fields on row select [April 27, 2026](#)
- ☒ Perform CRUD operations [April 27, 2026](#)

SQL

- ☒ INSERT [April 27, 2026](#)
- ☒ UPDATE [April 27, 2026](#)
- ☒ DELETE [April 27, 2026](#)
- ☒ SELECT [April 27, 2026](#)

5. CUSTOMER MANAGEMENT MODULE

- ☒ Customer form (name, contact, address) [April 27, 2026](#)
- ☒ Customer table [April 27, 2026](#)
- ☒ CRUD operations [April 27, 2026](#)
- ☒ Link customer to orders [April 27, 2026](#)

6. REPORT MODULE

USER INTERFACE

- ☒ Date filters (start/end) [April 28, 2026](#)
- ☒ Report table [April 28, 2026](#)
- ☒ Buttons:
 - ☒ Generate [April 28, 2026](#)
 - ☒ Export [April 28, 2026](#)
 - ☒ Print [April 28, 2026](#)

FUNCTIONALITY

- ☒ Generate:
 - ☒ Daily sales [April 28, 2026](#)
 - ☒ Monthly sales [April 28, 2026](#)
 - ☒ Product sales [April 28, 2026](#)
 - ☒ Compute totals [April 28, 2026](#)

EXPORT (password protected)

- ☒ CSV [April 28, 2026](#)
-
- ☒ PDF (does not support java, need to install)
 - ☒ DOC [April 28, 2026](#)
-

7. DATABASE DESIGN

TABLES

- ☒ Users [April 28, 2026](#)
- ☒ Products [April 28, 2026](#)
- ☒ Customers [April 28, 2026](#)
- ☒ Orders [April 28, 2026](#)
- ☒ Order Details [April 28, 2026](#)

SETUP

- ☒ Configure JDBC connection [April 28, 2026](#)
- ☒ Normalize tables [April 28, 2026](#)
- ☒ Use foreign keys [April 28, 2026](#)
- ☒ Handle exceptions [April 28, 2026](#)

8. EVENT HANDLING AND NAVIGATION

- ☒ Implement ActionListener [April 28, 2026](#)
- ☒ Use CardLayout (Used JTabbedPane) [April 28, 2026](#)
- ☒ Handle table clicks [April 28, 2026](#)
- ☒ Prevent unauthorized access [April 28, 2026](#)

9. UI/UX DESIGN

VISUAL DESIGN

- ☒ Consistent color scheme (2–3 colors) [May 3, 2026](#)
- ☒ Consistent fonts and sizes [May 3, 2026](#)
- ☒ Proper spacing and alignment [May 3, 2026](#)
- ☒ Use icons with labels [May 3, 2026](#)
- ☒ Avoid clutter [May 3, 2026](#)

LAYOUT

- ☒ Sidebar navigation [May 3, 2026](#)
- ☒ Highlight active menu [May 3, 2026](#)
- ☒ Group related components [May 3, 2026](#)
- ☒ Keep key actions visible [May 3, 2026](#)

USER EXPERIENCE

- ☒ Minimize clicks [May 3, 2026](#)
- ☒ Use clear labels [May 3, 2026](#)
- ☒ Add placeholder text [May 3, 2026](#)
- ☒ Enable keyboard navigation [May 3, 2026](#)
- ☒ Auto-focus inputs [May 3, 2026](#)
- ☒ Use ToolTipText [May 3, 2026](#)

ERROR HANDLING

- ☒ Friendly error messages [May 3, 2026](#)
- ☒ Highlight invalid fields [May 3, 2026](#)
- ☒ Confirmation dialogs [May 3, 2026](#)
- ☒ Success notifications [May 13, 2026](#)

FEEDBACK

- ☒ Loading indicators [May 13, 2026](#)
- ☒ Disable buttons during processing [May 13, 2026](#)

TABLES

- ☒ Sortable columns [May 13, 2026](#)
- ☒ Search/filter bar [May 13, 2026](#)
- ☒ Highlight selected rows [May 3, 2026](#)
- ☒ Align numeric values [May 13, 2026](#)

FORMS

- ☒ Logical grouping [May 13, 2026](#)
- ☒ Default values [May 13, 2026](#)
- ☒ Dropdowns instead of free text [May 13, 2026](#)
- ☒ Input validation [May 13, 2026](#)

NAVIGATION FLOW

- ☒ Smooth transitions [May 13, 2026](#)
- ☒ Back/Home buttons [May 13, 2026](#)
- ☒ Confirm exit [May 13, 2026](#)

ADDITIONAL FEATURES

- ☒ Dark mode [May 13, 2026](#)
- ☒ Animations (JavaFX) [May 13, 2026](#)
- ☒ Dashboard charts [May 13, 2026](#)

ACCESSIBILITY

- ☒ Readable fonts [May 13, 2026](#)
- ☒ High contrast colors [May 13, 2026](#)

- ☒ Keyboard accessibility [May 13, 2026](#)

10. SECURITY AND VALIDATION

- ☒ Input validation [May 13, 2026](#)
- ☒ Prevent SQL injection [May 13, 2026](#)
- ☒ Role-based access (Admin/User) [May 13, 2026](#)
- ☒ Exception handling [May 13, 2026](#)

11. TESTING

- ☒ Test login scenarios [May 13, 2026](#)
- ☒ Test order process [May 13, 2026](#)
- ☒ Test reports accuracy [May 13, 2026](#)
- ☒ Test database operations [May 13, 2026](#)
- ☒ Handle edge cases [May 13, 2026](#)

12. DEPLOYMENT

- ☒ Export as JAR file [May 18, 2026](#)
- ☒ Setup database configuration [May 18, 2026](#)
- ☒ Provide documentation:
 - ☒ User guide [May 18, 2026](#)
 - ☒ Installation guide [May 18, 2026](#)
- ☒ Backup database [May 18, 2026](#)

13. ADVANCED FEATURES (OPTIONAL)

- ☒ Barcode scanner [May 12, 2026](#)
- ☒ Receipt printing [May 12, 2026](#)
- ☒ Inventory alerts [May 12, 2026](#)
- ☒ Multi-user system [May 12, 2026](#)
- ☒ Sales analytics dashboard [May 12, 2026](#)

"Precision in Code, Passion in Design."

Figure 1. Login Form

The `'login.java'` file serves as the main authentication module of the system and was developed using Java Swing to provide a graphical user interface (GUI) for login, account registration, and password recovery. The interface uses customized components such as `'JFrame'`, `'JPanel'`, `'JButton'`, `'JTextField'`, `'JPasswordField'`, `'JComboBox'`, and `'JTabbedPane'` to create a modern dark-themed design with navigation panels, hover effects, and styled input fields. The program allows users to log in by entering their username, password, and role, while also validating whether required fields are completed and whether the username follows the allowed format. During authentication, the system connects to the MySQL database using JDBC and retrieves user records from the `'users'` table. Passwords are encrypted using the SHA-256 hashing algorithm before comparison to improve security. If the credentials match, the system stores the user session and redirects the user to the `'OrderingSystemMenu'` window. To further strengthen security, the program limits login attempts and locks the login form after three consecutive failed attempts.

The registration module allows users to create accounts by providing a username, password, security questions, birthday, and role. Before storing the information, the program performs several validations such as checking empty fields, validating birthday format, verifying username format, and ensuring password strength through regular expressions. Duplicate usernames are also prevented through database checking. Once validated, the account information is securely inserted into the database using prepared statements. In addition, the forgot password feature enables users to recover their accounts by answering security questions linked to their stored account information. If verified successfully, the user may reset their password, which is again encrypted before updating the database. Overall, the system demonstrates important concepts in software engineering including GUI development, database connectivity, user authentication, password hashing, validation, exception handling, and role-based access control, making it suitable for small-scale business systems and academic projects.

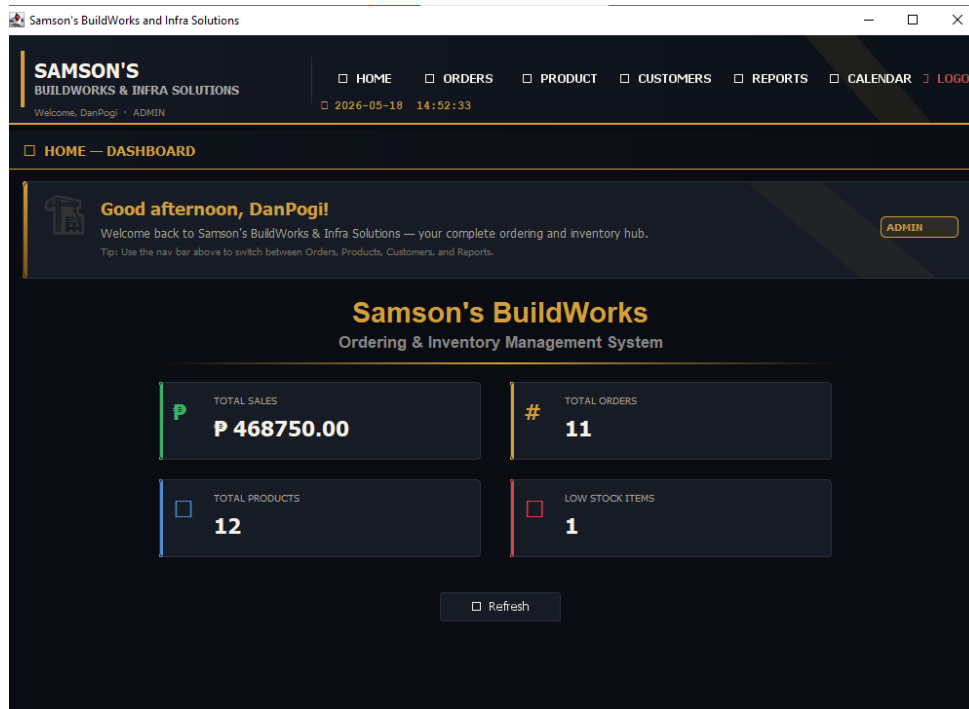


Figure 2. Dashboard Module

The Home Tab serves as the main dashboard of the Ordering and Inventory Management System, providing users with a quick overview of important business information upon login. The interface is designed using Java Swing components such as JPanel, JLabel, and JButton, with customized colors, gradients, and rounded panels to create a modern and professional appearance. A welcome section is displayed at the top of the dashboard, showing the company name, “Samson’s BuildWorks,” together with the system title. Decorative elements such as gradient dividers and styled cards improve the overall user experience and make the interface more visually organized. The dashboard layout is arranged carefully to ensure that important information is easy to locate and understand.

In addition, the dashboard contains four statistical cards that display key operational data: Total Sales, Total Orders, Total Products, and Low Stock Items. Each card includes an icon, title, and dynamically updated value loaded from the system through the loadDashboard() method. Different color indicators are also used to help users quickly identify categories such as warnings for low stock items or success indicators for sales information. A refresh button is provided so users can instantly update the displayed statistics without restarting the application. This feature demonstrates the use of event handling, graphical customization, and dynamic data presentation in Java, making the dashboard both functional and user-friendly for monitoring daily business operations.

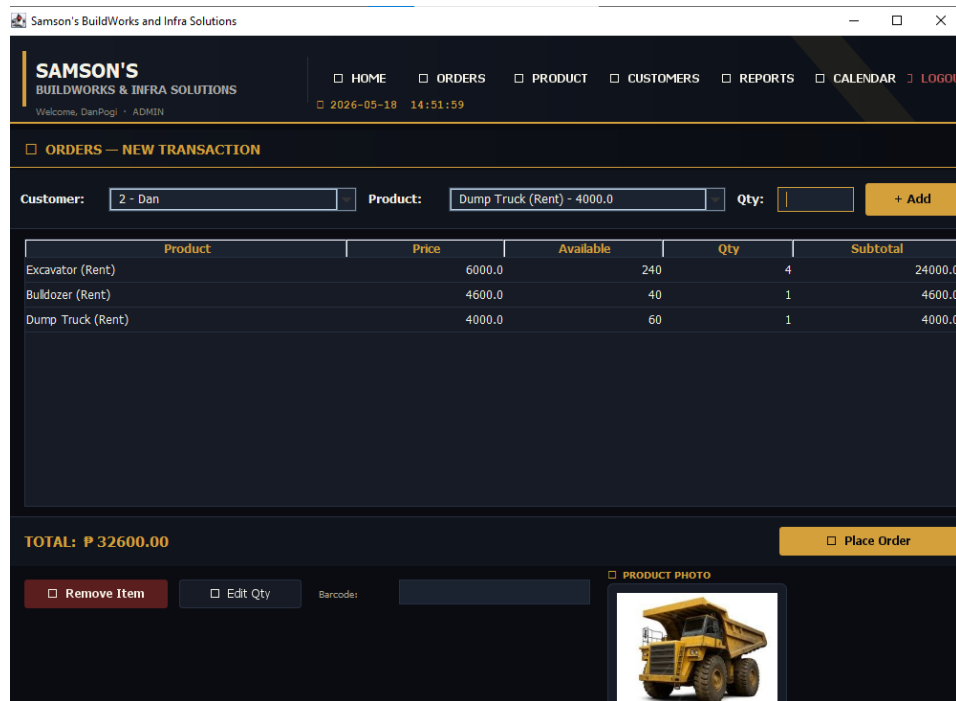


Figure 3. Ordering Management Module

The Orders Tab is responsible for handling customer transactions and managing the ordering process within the system. It allows users to select customers, choose products, enter quantities, and add items to a shopping cart using a user-friendly graphical interface built with Java Swing components such as JPanel, JTable, JComboBox, JButton, and JTextField. Product information is dynamically loaded from the database, including product names and prices, while customer records are displayed through a combo box selection menu. The system also supports barcode scanning, enabling faster product selection by automatically identifying items based on their barcode values. The cart table displays important transaction details such as product name, price, available stock, quantity, and subtotal, while the total amount is automatically calculated and updated whenever items are added, edited, or removed.

In addition, the Orders Tab contains several validation and inventory management features to ensure accurate transactions. The system checks whether the entered quantity is valid and prevents users from ordering quantities that exceed the available stock. Users may also edit or remove products from the cart before finalizing the transaction. Once the “Place Order” button is clicked, the program saves the order information into the database, inserts order details into a separate table, updates product stock quantities, and generates a unique order ID. A receipt is then automatically created and displayed using a formatted text area, with an option for receipt printing. Database transaction control using commit() and rollback() is also implemented to maintain data integrity in case an error occurs during processing. Overall, this tab demonstrates important programming concepts such as database connectivity, event handling, transaction management, inventory control, and real-time order processing in Java.

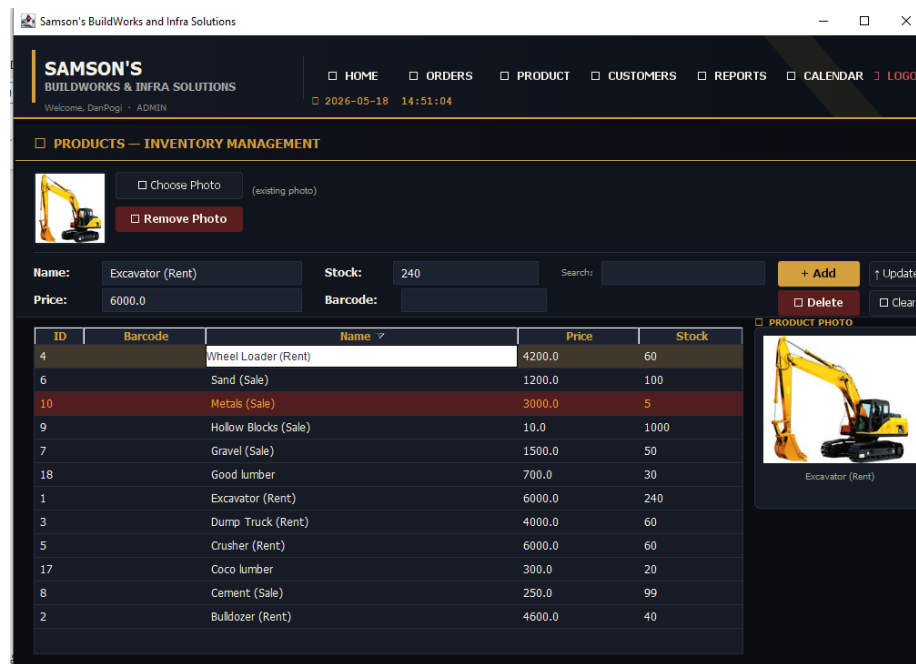
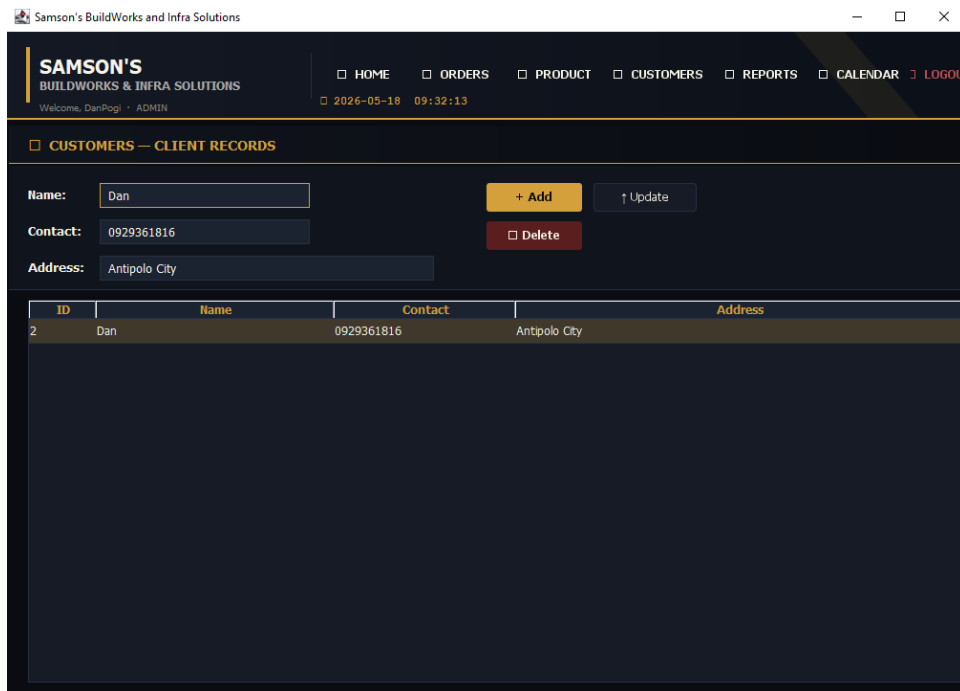


Figure 4. Product Management Module

The Products Tab is designed for inventory management and allows administrators to manage product records within the Ordering and Inventory Management System. The interface is built using Java Swing components such as JPanel, JTable, JTextField, and JButton, providing a structured and user-friendly environment for handling product information. Users can enter important details such as product name, price, stock quantity, and barcode through input fields located in the form panel. The tab also includes buttons for adding, updating, deleting, and clearing product records. A search feature is implemented to allow users to quickly filter and locate products based on their names, improving efficiency when handling large inventories. In addition, product information is displayed in a table that dynamically loads data from the database, including product ID, barcode, name, price, and stock quantity.

Furthermore, the Products Tab includes several validation and inventory monitoring features to maintain accurate records and improve system security. The system validates product prices and stock quantities to prevent invalid inputs such as negative values or empty fields. Low-stock items are automatically highlighted using custom table cell renderers, allowing administrators to easily identify products that require restocking. Administrative actions such as adding, updating, and deleting products are protected through an admin password verification process before changes can be applied. Database operations are performed using SQL queries with prepared statements to securely insert, update, search, and remove product records. After every modification, the product table and product selection combo boxes are refreshed to ensure that all displayed information remains updated. Overall, this tab demonstrates important programming concepts including CRUD operations, inventory control, database management, graphical user interface design, event handling, and input validation in Java.



The screenshot displays the 'CUSTOMERS — CLIENT RECORDS' section of the application. At the top, there is a navigation bar with links to HOME, ORDERS, PRODUCT, CUSTOMERS, REPORTS, CALENDAR, and LOGOUT. The current user is 'Dan Pogi' (ADMIN) and the date/time is '2026-05-18 09:32:13'. Below the navigation bar, the 'CUSTOMERS — CLIENT RECORDS' section is active. It features a form with three input fields: 'Name' (containing 'Dan'), 'Contact' (containing '0929361816'), and 'Address' (containing 'Antipolo City'). To the right of the 'Name' field are buttons for '+ Add' and '↑ Update'. Below the 'Contact' field is a 'Delete' button. Below the form is a table with the following data:

ID	Name	Contact	Address
2	Dan	0929361816	Antipolo City

Figure 5. Customer Management Module

The Customers tab is your go-to place for managing client records. From a straightforward form, you can add new customers, update their details, or remove them entirely — covering the basics like name, contact number, and address. The system won't let anything slip through incomplete either; all fields need to be filled out before anything gets saved. Browsing through existing records is just as smooth — click on any row in the table and the form fills itself in automatically, so making edits doesn't mean retyping everything from scratch. A few keyboard shortcuts are also built in to keep data entry moving without unnecessary clicks.

What makes the tab especially useful is how it connects to the rest of the system. Customer names feed directly into the Orders tab, so when someone's placing a transaction, pulling up the right client is just a quick selection away. Any updates you make here — new entries, edits, deletions — reflect immediately wherever customers are listed, so nothing ever feels out of sync. Confirmation prompts and success messages keep you in the loop throughout, and everything talking to the database does so securely, with proper safeguards against errors and unexpected issues.

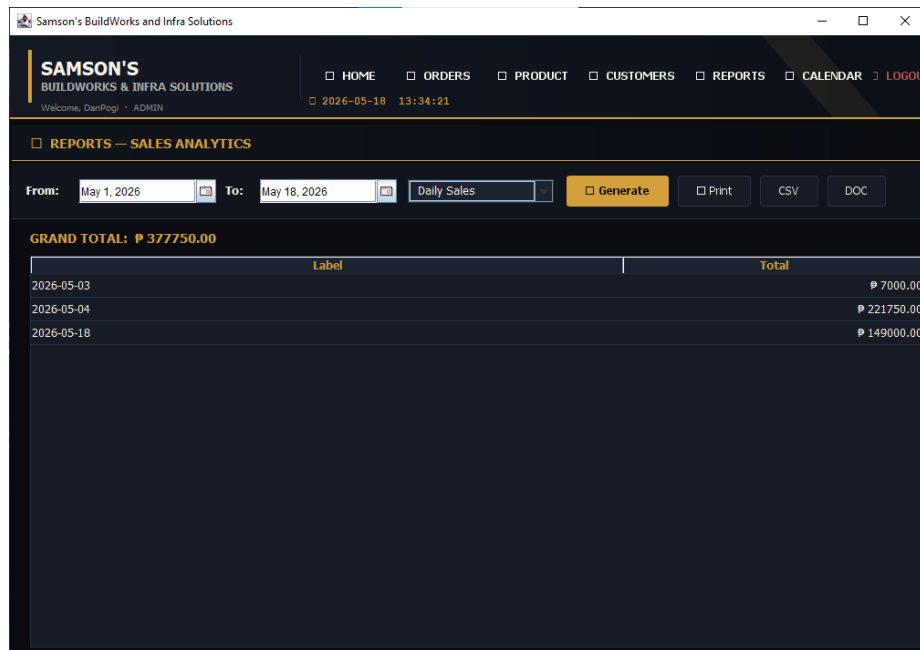


Figure 6. Reports Module

The Reports tab is where you go to make sense of your sales data. Whether you need a snapshot of today's activity, a monthly breakdown, or a product-by-product summary, you can pull up any of these reports simply by picking a date range. Results show up in a clean table with a grand total at the bottom, making it easy to spot trends and keep a pulse on how the business is doing — all without having to dig through raw data manually.

Beyond just viewing numbers, the tab gives you practical options for what to do with them. You can send a report straight to the printer, or save it as a CSV or DOC file for your records. To keep things secure, exporting requires an admin password — so sensitive business data doesn't end up in the wrong hands. The whole interface is laid out to feel like a proper analytics dashboard, with labeled buttons, helpful tooltips, and smooth error handling running quietly in the background.

What techniques and strategies did you apply to complete the system? What features does your system have that others do not?

This system was built using a structured, multi-class Java Swing architecture where authentication, session management, and the main application are separated into distinct components. Passwords are secured with SHA-256 hashing so plaintext credentials are never stored, and all database queries use PreparedStatements to prevent SQL injection. Login attempts are tracked and the account locks after three failures. Order placement uses database transactions with setAutoCommit(false) and batch execution, rolling back automatically if anything fails mid-process to prevent corrupt or partial records. Role-based access control is enforced through a Session class that carries the logged-in user's role across the application, and every meaningful action is written to an activity log table for auditing.

The UI is built entirely with custom paintComponent() overrides using Graphics2D, producing gradient backgrounds, gold accent bars, hazard stripe decorations, rounded buttons with hover states, and corner bracket markers — all without any external UI framework. Navigation is handled through a sidebar and top nav bar that programmatically switch a hidden JTabbedPane, keeping the tab headers invisible while maintaining clean panel separation. Input fields simulate placeholder text using FocusListener since vanilla Swing lacks native support for it, and field borders change color on focus or validation error to give immediate visual feedback.

On the functional side, the system supports barcode scanning to auto-select products, auto-generated date-stamped order codes, a live clock in the header updated every second via a Swing Timer, and a shopping cart with real-time stock validation, quantity editing, and item removal before checkout. After an order is confirmed, a formatted receipt is displayed and sent directly to a printer. The Reports tab generates daily, monthly, or product-level sales breakdowns with date range filtering, and exports to CSV or DOC files. The Products tab highlights low-stock items visually using a custom cell renderer, and the Forgot Password tab allows account recovery through security question verification.

Rubric: Java GUI Ordering System

Programmer: Dan Vanice D. Samson

Criteria	Excellent (4)	Good (3)	Satisfactory (2)	Needs Improvement (1)	Score
Login Module	Login works flawlessly, secure password hashing, proper session handling, error messages clear	Login mostly works, minor validation issues, session handled	Login works but lacks validation or security measures	Login incomplete or non-functional	
Dashboard Module	Fully functional dashboard, smooth navigation, dynamic panels, logout works	Minor navigation issues, most buttons functional	Basic dashboard, navigation limited	Dashboard missing or non-functional	
Order Management Module	Products load correctly, cart updates in real-time, totals computed, orders saved to DB, stock updated	Most functions work, minor UI/logic issues	Basic add to cart works, some DB operations missing	Incomplete or non-functional	
Product Management Module	Full CRUD operations, proper validations, table displays all products	Minor errors in CRUD or validation	Basic CRUD with missing validation	Incomplete or non-functional	
Customer Management Module	Full CRUD operations, linked to orders, validation applied	Minor errors in CRUD or linking	Basic CRUD, partial functionality	Incomplete or non-functional	
Reports Module	Reports generated accurately, filters work, export to CSV/PDF, totals computed	Minor inaccuracies, exports partially working	Basic report generation	Incomplete or non-functional	
Database Design	Proper normalized tables, foreign keys implemented, JDBC working without errors	Minor issues in table design or DB connection	Basic DB connection, some tables incomplete	Poor or missing database design	
UI/UX Design	Clean, professional, consistent design, smooth flow, accessible, keyboard navigation, proper feedback	Mostly consistent design, minor flow issues	Basic design, lacks some UX principles	Poor UI/UX, confusing layout, no feedback	
Event Handling and Navigation	All buttons and panels responsive, CardLayout/Scene switching works smoothly, unauthorized access prevented	Minor event issues, mostly functional	Basic events work	Non-functional or missing	
Security and Validation	Proper input validation, SQL injection prevented, role-based access enforced, exceptions handled	Minor validation/security gaps	Basic validation, some vulnerabilities	Poor or missing security	
Testing	All modules tested, edge cases handled, no major bugs	Minor bugs, mostly stable	Some testing done, multiple bugs	Untested or non-functional	
Deployment	JAR file runs without errors, DB configured, documentation complete	Minor deployment issues, documentation mostly complete	JAR or DB setup incomplete	Non-functional or missing	
Advanced Features (Optional)	All features implemented (barcode, receipt printing, alerts, analytics)	Some features implemented	Minimal advanced features	No advanced features	
TOTAL SCORE					

DOCUMENTATION:



(exporting it as a JAR file)



(testing it as an costumer)



(testing it as an admin)



(final testing/running)

QR CODE FOR SOURCE CODES:

